



Second Year Group Project Guidelines

**Bachelor of Science (Honours) in Computer Science Degree
CSCI 23072 Group Project**

Faculty of Computing and Technology
University of Kelaniya

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Structuring your project report

Report is the most important document for this course and it should be in the form of a document of not less than 5000 (excluding the appendices). It is recommended to use any text editor to compose the group project. When the students are composing the report, the following format may help to make high-quality project report.

(Font type: Times New Roman, Font size: 12 Pt, Line Spacing: 1.5)

- ❖ Title Page
 - Project title
 - Group number, student details and name of the supervisor
 - Statement that the report was submitted in partial fulfillment of the requirements of the above module in the B.Sc. (Hons) degree in Computer Science
 - Faculty and University
 - Month and Year
- ❖ Abstract (Maximum one page)
- ❖ Acknowledgement
- ❖ Table of Content
- ❖ List of Figures/ Tables
- ❖ List of Abbreviations
- ❖ Main text
 - Introduction
 - Motivation for the project
 - Project Objectives
 - Scope of the Project
 - Limitations
 - Background
 - Problem Identification
 - Literature Review
 - Methods and Tools
 - Design
 - Overall Design
 - Selected sections to be implemented (with justification)
 - Detailed design of the implemented section (description and diagrams)
 - Implementation
 - Technologies used
 - User Interface Design
 - Reports Generated
 - Discussion and Conclusion
 - Discussion on the process followed during the project
 - Conclusions drawn

- ❖ Future Work
- ❖ References
- ❖ Appendices (If available)

The brief guide for each component of the project report is as follows:

Acknowledgement

The students should acknowledge the assistance given by their supervisors, and any other person or organization that has helped them in the planning, conduct, analysis or reporting of the project.

Abstract

This is an outline of the study question, aims and objectives, background literature, methods, results, key conclusions and recommendations. This should be 250-300 words long and should be very clear and easy to follow. It is recommended to include quantitative or qualitative outcomes which are drawn from the research in brief.

Introduction

This section is basically used to introduce the research problem or knowledge gap, outline of aims and objectives and the motivation to conduct the particular study.

Literature Review

This section is used to demonstrate student's investigation ability in particular research problems considering different potential dimensions. Students are expected to conduct thorough literature review on current work of the research problem and structure the review with a smooth flow as to make easy to understand by the reader. It is required to cite appropriately when the students draw conclusions out of current works.

Methods

This section includes the methodology of the research. It can be a form of the study design, the study population, sampling frame and numbers, sampling method, survey design, survey or data collection instruments, protocol for obtaining data, ethical issues and how they are addressed, data management and analysis methods and statistical analysis and tests.

Results

This section includes the entire outcomes from the research project. This section includes qualitative or quantitative measures in form of figures, graphs and tables. All tables, figures and graphs, should be numbered consecutively throughout the whole report and labeled with a clear and concise descriptive title. Students should pay special attention about the effective presentation techniques while drawing figures and graphs.

Discussion

The proper interpretation of the findings presented in this section. Any limitations in the research methodology should also be referred to here. Examiners expect students to acknowledge these limitations as an integral part of the evaluation of the project.

Conclusion

This section summarizes the key results and the conclusions that the student can draw from these results. It also needs to reflect what the initial project aims and objectives were.

Future Work

It is good research practice to make recommendations or to suggest directions for further research or actions as a result of the project findings.

References

As a professional researcher, students are expected to avoid plagiarism and if any other researcher's work is considered or based in the project, it is required to properly refer to these works with original researcher details. This is a list of all such references and sources which were used in the literature review, methodology and discussion. The references can be books (monographs), journal articles, letters, abstracts, patents, conference and symposium papers, media articles, and any form of published literature or comment both offline and online forms. It is important that every claim of scientific fact the students make is supported by a valid, relevant, accessible reference, and that every idea or argument, and every verbatim quotation or paraphrase of someone else's work, is correctly attributed to its source. Students are advised to follow IEEE Reference style.

Appendices

Students may include the external resources which were used to conduct the research in this section. Those resources can be any questionnaires or evaluation instruments used, covering letters, participant information and ethics approvals, statistical formulas, mathematical proofs or additional explanations.

Requirement Analysis and Specification

This includes the analysis, purpose, scope, functional and nonfunctional requirements, software and hardware requirements of the project. It should document all the analysis and design models considered and justify the design decisions. In addition to this, it also contains the information about environmental conditions required, safety and security requirements, software quality attributes of the project etc.

Design and Implementation

This section should describe the complete design processes and the technology stack of the project. The hardware and software requirements of the project along with justifications for each choice is typically expected.

Testing and Evaluation

This section should describe how the developed system is tested and how the final system is evaluated. It may include feedback from users/ experts' reviews.

Deployment

This section includes how the software is deployed, the issues faced during deployment and how to solve the issues.

Important:

Avoid plagiarism – it is an offence

You will find valuable information on plagiarism on this site:

<https://www.ox.ac.uk/students/academic/guidance/skills/plagiarism?wssl=1>

Preparing Oral presentation

Students are notified the date and venue for the oral presentation (physically) through the course coordinator after submitting the final version of the report approved by the supervisor.

This presentation includes two sessions: students presenting session which stands for 15 minutes and panel questioning session which stands for 5 minutes. Students should be vigilant about the time constraints when creating the presentation slides and students will be penalized if particular group could not finish the presentation in the allocated time.

The presentation slides can be organized in the same flow that has been used for the project. In the first slide, title, supervisor name and students index numbers should be clearly shown. Students should pay special attention to the best practices for making an effective presentation. A demo is mandatory for development/implementation-based projects.

The recommended structure for the presentation is as follows:

- ❖ First slide with title, supervisor name and students of the group along with index numbers
- ❖ Introduction
 - Background of the Problem
 - Scope and Objectives
 - Methods and tools
- ❖ Design
 - Overall Design
 - Detailed design of implemented sections
- ❖ Implementation
 - Technologies
 - User Interface
 - Testing and evaluation
- ❖ Discussion and Conclusion
- ❖ Future Work
- ❖ Table describing the Individual Contribution of group members
- ❖ References